

Ashton Honnecke

Senior Full-Stack Engineer • Python • TypeScript/React • AWS • AI-Native (agents, MCP, vision)

published_cv@cxcs.us • <https://ashton.honnecke.us> • <https://www.linkedin.com/in/ashtonhonnecke/> • <https://github.com/ahonnecke/> • Denver, CO

Highlights

- ~25 years shipping production systems across Python, TypeScript, and AWS — from Linux internals to cloud architecture.
- Builds AI artifacts, not just AI usage: a Claude vision QC gate and a Model Context Protocol (MCP) server (Alt Vault), an email-negotiation agent (Hagglebot), and contract generation on a self-hosted LLM (Toy Contractor) — shipped tools, not demos.
- Full-stack by default: Python/AWS backends and Next.js/React/TypeScript frontends, taken from design doc through shipped UI.
- Ships solo and at team scale: greenfield AI-native SaaS, 7M+ rows/day ingest, 100 GB/night data platforms, and extremely high-scale public-facing web properties.
- Speaker: PyColorado — Cleaning Up Your Python Environment: Superfund Site.
- Speaker: PyDEN — The Value of Linting.

Experience

CrewCapable AI - Chief Technology Officer (2025-Present)

Subject Matter Expert: TypeScript, Python, Next.js 15, React 19, AWS CDK, Lambda, Step Functions, Postgres/Supabase, LLM extraction, Zod, Vitest, LocalStack, EDI X12

- Greenfield AI-native SaaS for corrugated packaging purchase-order intake: OCR + LLM extraction + human-in-the-loop review + automated ERP/EDI export; single-handedly productionized from a proof of concept to a horizontally scalable cloud deployment in under 10 months.
- Serverless AWS architecture on CDK: Lambda functions and a Step Functions workflow orchestrating OCR/extraction/reconciliation; event-driven integration across S3, EventBridge, SNS, Secrets Manager.
- Split infrastructure into four independently deployable CDK stacks (core backend, audit, ERP, management) to isolate blast radius and enable per-tenant rollout.
- Next.js 15 / React 19 frontend over a typed API layer; Postgres (Supabase) with row-level security on every tenant-scoped table.
- Agentic LLM document-extraction pipeline (Gemini) with Zod-typed schemas; immutable audit trail separating raw model output from human-reviewed values.
- Document-processing state machine handling PDF, DOCX, image, and email-attachment intake with explicit failure and timeout paths.
- Test suite spanning Vitest unit tests, Testing Library, LocalStack integration, and ERP golden-file contract tests; CI covering type checks, migration linting, and ERP regression.
- Strict end-to-end type safety: branded domain types, Zod validation at every boundary, generated Supabase types; Biome + ESLint + Ultracite enforced in CI.
- Forward-only Postgres migrations run through a proper SDLC with queued migration promotion: promoting a build to staging or prod applies its migrations automatically, with linting in CI to prevent drift.
- Per-Lambda CloudWatch alarms, state-machine failure alerts, commit-SHA logging on Lambda cold start, /api/version endpoint for incident-to-revision traceability.

- AWS Secrets Manager + per-Lambda IAM scoping + RLS on every customer-data table — no shared service accounts across tenants.
- Production integrations: Amtech DAT (SFTP), QuickBooks, outbound X12 850 EDI; Stripe billing, Supabase Auth with Google OAuth, PostHog analytics, SES-based email intake.
- Turborepo / pnpm monorepo spanning web, backend, and shared packages with LLM-optimized single-concern file markers.

Panasonic North America - Lead Cloud Engineer (2020-2025)

Subject Matter Expert: Python, boto3, Bash, AWS (Lambda, SQS, SNS, Fargate, S3, CloudWatch, Cost Explorer), SonarQube, Docker, V2X/OBU telematics, SAE J2735 (DSRC), edge devices

- Chaired cross-team best-practices workgroup (~50 engineers) on OO, SOLID, code quality, and security; mentored developers across the org.
- Authored ADRs for event-driven/serverless and long-running serverful (Fargate) cloud-native architectures.
- Authored end-to-end incident playbooks: classification → response → resolution → blameless post-mortem.
- Built a near-real-time OBU ingress pipeline on Lambda/SQS/SNS with horizontal scale; worked with SAE J2735 DSRC messages in the data path; provisioned for throughput headroom not exercised in production.
- Shipped V2X messaging deployment for a Utah snowplow fleet; worked on edge devices, packet sniffing, and hardware↔cloud integration.
- At an internal hackathon (3-person team, one day), built an iOS and Apple Watch app that drove physical EV-charging hardware in real time — native phone and wrist control of the charge-management controller.
- Designed and released internal developer tooling as open-source on PyPI: hot Lambda filesystem mounting (Consolo), CLI SNS inspection (Snifter), Jira↔Git forge integration (Jolly Brancher).
- Delivered artifact build pipeline reducing build time and cost.
- Identified and decommissioned unused AWS resources via Cost Explorer and tagging discipline → up to ~30% infrastructure savings.
- Drove self-hosted SonarQube rollout across the org → >50% unit-test coverage gains across multiple languages.

Digital Assets Data - Sr. Software Engineer / Blockchain Tech Lead (2018-2020)

Subject Matter Expert: Python, Django, React, Pandas, Docker, AWS (Lambda, API Gateway, S3, CloudWatch, ECS), Google BigQuery, CircleCI, BTC/ETH nodes, smart contracts

- Built Python microservices to extract on-chain BTC/ETH data directly from cloud-hosted nodes (low-latency ingest; pandas; scheduled CloudWatch Events).
- Implemented Lambda + API Gateway microservices for cryptographic decoding at very high throughput.
- AWS ingestion pipelines on S3, CloudWatch Events, Lambda, and ECS — processed 100 GB nightly with individual files up to 30 GB.
- Worked with Google BigQuery to extract, normalize, and ingest Bitcoin and Ethereum datasets.
- Implemented QR-code TOTP multi-factor authentication in Django/React; hardened and containerized blockchain nodes across per-env configurations.
- “Fan-out” CircleCI workflows to parallelize PR tests; standardized unit testing + linting; containerized every stack with Docker.
- Automated external-library dependency PRs with integration checks against external APIs in the PR test flow only.
- Tech lead with 1 direct report (blockchain team of 2 at peak); owned on-call playbooks (VictorOps, OpsGenie); performed many interviews.
- Built centralized unhandled-exception capture/logging and uptime monitoring across multiple projects and languages.

- Restructured deploys into a single "push-button" script; wrote and distributed git hooks for linting, typing, and stylistic cohesion.

Havenly - Senior Software Engineer (2015-2017)

Subject Matter Expert: PHP, CakePHP 2/3, Linux, Git, SQL, queued-ingest systems, data modeling

- Most senior engineer on the data-ingest initiative within an ~8-person team; designed and shipped a greenfield, horizontally scalable queued data-feed system: 7M+ rows/day from 25+ sources in 15 formats.
- Built decoupled pipeline: raw retention (traceability/replay) → transform/validate/store → push to external API with faceted fuzzy search.
- Authored CakePHP2 → CakePHP3 migration plan; refactored into OO Components and Behaviors (CakePHP's mixins/traits) to enable encapsulation and unit testability.

Colorado State University Global Campus - Systems Architect (pro tem) (2014-2015)

Subject Matter Expert: Linux, PHP, SQL, data modeling, SDLC, pentesting, Git, SSH

- Covered a vacated architecture-leadership role at CSU Global (director-level scope); led 1 engineer, owned architecture and SDLC for student portal, course registration, and transcript systems.

Pixelstube Consulting - Founder (2009-Present)

Subject Matter Expert: Next.js, React, TypeScript, Postgres, Python, Docker, Fly.io, Linux, SQL, SDLC, Git, LLM extraction pipelines

- Founder of a long-running independent consultancy (profitable when actively engaged); broad engagements across web, mobile, data, and DevOps.
- Solo-delivered a multi-tenant HVAC load-calculation SaaS (Next.js 16 / React 19 / TypeScript, Postgres) from RFP to production for an engineering consultancy, replacing an Excel/Bluebeam workflow with an auditable web app.
- Architected an explicit IP boundary (client's proprietary calc engine behind a validated Python interface), a typed API client statically verified against server route contracts, an AHRI-catalog ETL into Postgres with SEER-tiered equipment ranking, and an immutable audit trail tying every result to its inputs and calc-engine version; deployed on Fly.io via GitHub Actions.
- Earlier: an LLM-driven image-to-structured-data pipeline for menu ingestion (picture → structured records).

Photobucket - Software Engineer → Senior Software Engineer (2005-2009)

Subject Matter Expert: PHP, MVC frameworks, PayPal API, credit-card payments, trust & safety tooling, AJAX, CI, MySQL

- W-2 engineer at Photobucket during its peak-traffic era (widely reported as top-50 global web property, ~100M users); promoted SWE → Sr SWE based on full ownership of internal tooling.
- Built trust & safety reporting pipeline: tool that packaged offensive content and transmitted to the National Center for Missing & Exploited Children (NCMEC); contributed to multiple predator arrests via proactive law-enforcement integration.
- Owned payments (PayPal + credit-card processing, reconciliation) and account-management tooling used for legal/compliance archival.
- Designed and hand-rolled an ETL process and tool to perform arbitrary data migrations without locking tables.
- Created a Propel-like model code-generation system with unit tests; migrated raw PHP pages to an MVC framework for internal tools.
- Introduced CI and unit testing into the team's workflow; mentored engineers on design, flow, and integration.
- Built an AJAX-based status/exception display system for consistent UX, surfacing caught DB exceptions.

Skills

- Software development across multiple languages, frameworks, and stacks (~25 yrs).
- Linux/Unix administration (~20 yrs); DevOps (cloud & colo) (~10 yrs).
- Startups (founder & early-stage) (~15 yrs), including profitable exits or profitability (~10 yrs).
- AI/LLM-native engineering (agentic extraction pipelines, Claude vision QC, MCP servers, agent-directed development, multi-provider orchestration) (~1 yr production, ongoing).
- Blockchain (ETH/BTC nodes, data engineering) (~2 yrs).

Speaking

- [Cleaning Up Your Python Environment: Superfund Site](#)
 - Slides and Code: <https://github.com/ahonnecke/pycolorado2019>
 - Video: https://www.youtube.com/watch?v=cpseEHA_haA
- [The Value of Linting](#)
 - Materials: <https://github.com/ahonnecke/linting-talk>

FOSS Utilities

Alt Vault

- Event-sourced vault intake pipeline for a physical-asset marketplace, built around one metric: time from item received to live. Live demo + full source.
- ScanGate — a Claude vision QC gate that catches glare, misalignment, blur, or a manifest mismatch before an item goes live.
- VaultTrace — a Model Context Protocol (MCP) server over the event log, answering “where is this item and what happened to it” in one sentence. Next.js / TypeScript / Postgres.
- <https://github.com/ahonnecke/alt.vault>

Hagglebot

- AI agent that negotiates car prices over email: FastAPI + async Postgres pulls dealer mail over JMAP, an LLM extracts offers, and it drafts counter-offers across dealerships. React frontend, Stripe billing, deploy-on-push CI.
- <https://github.com/ahonnecke/hagglebot>

Toy Contractor

- Generates and iteratively refines legal contracts from plain-English prompts on a fully self-hosted Mistral model via Ollama — no external API. FastAPI + Redis.
- https://github.com/ahonnecke/toy_contractor

Consolo

- Hot-mount Lambda filesystems for rapid iteration; “hot reloading.”
- <https://pypi.org/project/consolo/>

Snifter

- Fast SNS topic/message inspection from the CLI.
- <https://pypi.org/project/snifter/>

Jolly Brancher

- Git/Jira helper: find tickets, create matching branches, enforce branch hygiene.
- <https://pypi.org/project/jolly-brancher/>

Education

Master of Business Administration — University of Denver (2002-2004)

- Emphasis: Accounting

Bachelor of Computer Science — University of Denver (1999-2003)

- Minors: Mathematics, Philosophy